



DPURandomOscillator

This DPU creates a random-controlled signal in a predetermined waveform.

1. Inputs:

<i>In</i>	Input
<i>On</i>	Flag to switch the DPU On/Off

2. Dynamic parameters:

<i>Signal</i>	Waveform.
<i>AmplitudeMin</i>	Minimum amplitude of the signal
<i>AmplitudeMax</i>	Maximum amplitude of the signal
<i>PeriodMin</i>	Minimum period of the signal
<i>PeriodMax</i>	Maximum period of the signal
<i>FadeInTime</i>	Fade-in time of the signal
<i>FadeOutTime</i>	Fade-out time of the signal
<i>AlternationProbability</i>	Probability for alternation of the algebraic sign

3. Static parameters:

<i>SignalType</i>	Waveform (NOP,RandomTrapezoid,RandomSquare or RandomTriangle).
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4. Outputs:

<i>Out</i>	Output of the DPU
<i>AmplitudeAkt</i>	Current amplitude
<i>PeriodAkt</i>	Current period
<i>State</i>	State of the internal automaton

5. Functional principle:

The following waveforms are available:

- NOP (*Signal* = 0) no signal
- **RandomTrapezoid:** (*Signal* = 1) trapezoidal waveform
- **RandomSquare:** (*Signal* = 2) rectangular waveform
- **RandomTriangle:** (*Signal* = 3) triangular waveform

The waveform is either set by specifying *SignalType* or by setting the input of *Signal*. The DPU generates a signal with a random, equally distributed amplitude in the range between *AmplitudeMin* and



AmplitudMax. Afterwards the amplitude is multiplied by a sign multiplier. By drawing a random number between 0.0 and 1.0 it is determined, whether the sign multiplier is alternated. If an *AlternationProbability* of 1.0 is specified, the sign multiplier is always alternated. Parameters *FadeInTime* and *FadeOutTime* determine the rise and fall time of a trapezoidal waveform.

6. Example:

The following script is a minimal example with which DPURandomOscillator can be used:

```
DPUKeyboardState keyboard
{
    Computer = {LOCALHOST};
    Index = 1;
};

DPURandomOscillator randomOscillator
{
    Computer = {LOCALHOST};
    Index = 1;
    In=1;
    Signal=3;
};
Connections =
{
    keyboard.R -> randomOscillator.On
};
```

While holding “R” on the keyboard, the DPU will generate a randomized triangular (*Signal* = 3) waveform around the value 1 (*In*=1).